Komendra Sahu

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Summary

GenAI / RAG chatbot developer with a live production deployment for the Chhattisgarh state government and peer-reviewed research in NLP and information retrieval. Specializes in retrieval-augmented generation, semantic search, and end-to-end chatbot systems (ingestion → vector retrieval → LLM response → deployment).

Core Skills

GenAI / RAG: LangChain, Retrieval-Augmented Generation, RAG Fusion, Semantic Search, Vector Embeddings, Hybrid Dense-Sparse Retrieval, LLM APIs, Prompt Engineering
Vector & Data: Qdrant, PostgreSQL, PDF/OCR Processing Pipelines
Backend & APIs: Python, Flask, FastAPI, Node.js, Express.js, REST APIs, Server-Sent Events (SSE)
ML/NLP: Scikit-learn, PyTorch, Statistical Modeling, NLP
Tools: Git, Docker (basic), Linux, Pandas, NumPy

Featured Project — Production Deployment

RTI Assistant — Legal RAG & PIO Advisory System *Live client work, CG Government*
Python, Flask, JavaScript, Node.js, Qdrant, PostgreSQL, BGE-M3, LLM APIs [Repo – VERIFY LINK]

- Built an end-to-end legal-intelligence chatbot for Chhattisgarh RTI (Right to Information) workflows: semantic RAG over legal documents, officer-directory search, and automated PIO (Public Information Officer) advisory generation.
- Implemented Qdrant-based legal document retrieval and a PostgreSQL-backed PIO/FAA lookup system, with full CIC/CGSIC precedent expansion and clickable source-PDF citations for every generated answer.
- Designed a structured advisory pipeline that extracts the RTI issue, maps it to applicable Act provisions, retrieves relevant commission decisions, and generates evidence-bound responses with verification safeguards against hallucination.

Experience

CMIT Fellow *Present*

- Developing GenAI / RAG chatbot solutions for Chhattisgarh government departments, from data ingestion through production deployment.
- Working directly with real-world, unstructured government datasets under practical infrastructure and compliance constraints.

Machine Learning Intern — Unified Mentor *Jan 2026 – Apr 2026 — Remote*

- Built a student segmentation and course-recommendation system using unsupervised learning; applied PCA and K-Means (Silhouette Score 0.633).
- Built a hybrid recommender combining SVD-based collaborative filtering with cluster-level learner insights for the EduPro platform.

Other Projects

- **Financial Text Complexity Analytics (NLP & ML)**
  Python, FastAPI, React, Random Forest [Github]
  Built the Corporate Communication Text Complexity Index (CCTI) over 200K+ SEC filings and trained a Random Forest model to link textual complexity to stock returns, outperforming standard sentiment-based baselines.
- **Real-Time Traffic Monitoring System (Edge–Cloud ML)**
  Python, YOLOv8, Flask, React, CUDA [Github]
  Built an edge–cloud vehicle-detection system with YOLOv8; improved inference throughput via asynchronous, rate-limited communication; deployed REST APIs with a live React monitoring dashboard.

Research & Publications

- **Dynamic Query Handling with RAG Fusion for PDF-Based Knowledge Retrieval Systems**
  Published, IEEE Xplore, 2025 [Link]
  Proposed a retrieval-augmented framework for efficient document understanding and query handling over unstructured PDFs.
- **Optimization of 100M-Scale Click Fraud Detection Pipelines on Resource-Constrained 2-Node YARN Clusters**
  Accepted, ICST 2026 *[VERIFY LINK – currently duplicates the paper above, fix before sending]*
  Designed an optimized Apache Spark/YARN pipeline to process 100M click records on a two-node cluster using eager checkpointing and hardware-aware memory configuration (ROC-AUC 0.9058, Random Forest).
- **Nonlinear Effects of Text Complexity in Corporate Disclosures (CCTI)**
  Accepted, ICTIS 2026 [Link]
  Developed the CCTI index and showed nonlinear relationships between textual complexity and stock returns using Random Forest models.
- **Shift-Aware Meta-Reinforcement Learning for Robust Auto-Scaling in Serverless Clouds**
  Accepted, ICTIS 2026 [Link]
  Proposed a KL-divergence-based shift-detection mechanism integrated with Meta-PPO for adaptive autoscaling under dynamic workloads.

Education

M.Tech in Data Science & Artificial Intelligence *2025–2027*
IIIT Naya Raipur, India **CGPA: 9.16**

GATE 2025 (Computer Science) **Score: 585**

B.Tech in Computer Science & Engineering *2021–2025*
Government Engineering College, Raipur